Quantum Glitchy Period Finding

Status report and open problems

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Compiled October 2026 from the 2019 poster, simulation code and logs in this repository

Abstract. Quantum period finding assumes an exactly periodic oracle. We study a periodic sequence corrupted by k insertions or deletions ("glitches"). Each glitch shifts every later entry, so the corrupted function disagrees with every periodic function on a constant fraction of its domain. This note records the state of the 2019 project: the idealized model used throughout, the exact measurement distribution, the one-glitch theorem (probability near a good outcome at least 0.447 times ideal), the multi-glitch conjecture, and the numerical evidence. The 2019 materials (poster, code, logs) contain no written proofs. Material marked new in this note was reconstructed or added while compiling this note. It includes an identification of the one-glitch constant as $8/(8+\pi^2)$, a Parseval reformulation of the multi-glitch problem, a crude bound with quadratic radius, and a family of configurations showing that the conjectured constant $1/2$ would be asymptotically sharp. Separately, an exact residue-counting argument proves total probability at least $1/5$ at the ideal outcomes themselves after two literal glitches, without resolving the stronger per-outcome conjecture. A list of open problems closes the note.

How to read the status labels. from 2019 work the statement or formula appears in the 2019 poster, code or logs. 2019 claim, proof not on file stated on the 2019 poster; no proof is in the repository. new in this note added while compiling this note. proof below a complete argument is given here. proof sketch only the argument is outlined and the details still need to be checked. numerical evidence only supported by computation only. conjecture open. Statements marked new in this note were checked numerically with `uv run glitchy.py check` (Appendix A). They were prepared with AI assistance and have not been independently refereed.

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1 Summary of the current state

- **Model.** All 2019 analysis and simulation use one idealized *block-shift model* (Definition 1). In it every glitch sits on a period boundary and shifts all later copies of the period by ± 1. Its measurement distribution has an exact closed form (Lemma 2).

- **One glitch.** The poster states that for $r \mid N$ the probability of an outcome within distance 1 of a good outcome is more than 0.447 times the ideal probability. In the continuum limit we reconstruct the argument (Theorem 8). The neighbourhood ratio is at least $1 - 4u(1-u) + (8/\pi^2)\sin^2(\pi u)$, and Jordan's inequality turns this into $8/(8+\pi^2) = 0.4477...$, matching the poster's constant. The sharp value is 0.6400.
- **Several glitches.** The poster's "gruesome expression" is exactly the continuum neighbourhood mass (Section 4.3). The 2019 target was to prove that it is at least 1/2 (Conjecture 5). Exhaustive simulations for $k = 2, 3, 4$ give worst cases 0.772, 0.614, 0.689 (Table 2). The 2019 grid search for $k = 4$ had found only 0.723. The worst good outcome is $c = r/2$ for $k = 1, 3$ but not for $k = 2, 4$.
- **Added in this note.** (a) The neighbourhood mass is 1 minus the high-frequency energy of a unimodular step function, with an explicit formula in terms of jumps (Proposition 6). (b) Radius $L = \lceil 4(k+1)^2/\pi^2 \rceil$ already gives mass $\geq 1/2$ (Corollary 7), so the open problem is to reduce the radius to $O(k)$. (c) Error bounds for passing between the discrete and continuum models (Lemma 4, Proposition 10). (d) A packed family whose mass at radius k decreases towards 1/2, reaching 0.521 at $k = 200$ (Section 8.1). (e) Any bound uniform in c needs radius at least $(k+1)/2$ (Proposition 9).
- **Two glitches, total mass.** For two literal insertions/deletions at arbitrary positions, with arbitrary inserted symbols, the total probability at the exact ideal outcomes is at least $r/(3r+2) \geq 1/5$ (Theorem 3). This also bounds the probability within distance 2. Unlike the conjecture, it is an averaged statement, not a lower bound near each good outcome.
- **Not done.** No multi-glitch theorem uniform in c at radius $O(k)$. No period-recovery theorem (Section 9). No analysis of $r \nmid N$. No public follow-up paper or manuscript exists as of October 2026 (Section 10, item 8).

2 Standard period finding

Let $f : \mathbb{Z}_N \to W$ have period r, with $r < \sqrt{N}$. For most of this note we also assume[1]

$$r \mid N, \quad M \coloneqq N/r, \quad f(x) = w_{x \bmod r} \text{ with } w_0, ..., w_{r-1} \text{ distinct.} \tag{1}$$

The algorithm prepares $N^{-1/2} \sum_x |x\rangle|f(x)\rangle$, applies the quantum Fourier transform to the first register and measures. Outcome y appears with probability

$$P_f(y) = \frac{1}{N^2} \sum_{w \in W} \left| \sum_{x : f(x) = w} e^{2\pi i x y / N} \right|^2 . \tag{2}$$

For the ideal function, value w_a is taken on $\{a + mr : 0 \leq m < M\}$, and the inner sum is $e^{2\pi i a y/N} \sum_{m=0}^{M-1} e^{2\pi i m y/M} = M e^{2\pi i a y/N} \mathbb{1}[M \mid y]$. Hence

$$P_0(y) = \frac{1}{r} \cdot \mathbb{1}[M \mid y], \quad G \coloneqq \{cM : 0 \leq c < r\}. \tag{3}$$

From a good outcome $y = cM$ we get $y/N = c/r$. Reducing the fraction gives $r/\gcd(c, r)$, and $\Pr[\gcd(c, r) = 1] = \varphi(r)/r = \Omega(1/\log\log r)$. When $r \nmid N$, the peaks sit at the integers nearest cN/r. Continued fractions then recover c/r because $r^2 \leq N$. This is where the assumption $r < \sqrt{N}$ is used.

[1] The poster's figure 5 treats $r \nmid N$; nothing is proved there.

3 The glitch model

3.1 Informal picture

An *insertion* adds one symbol to the periodic sequence and shifts everything after it by one place. A *deletion* removes one symbol and shifts everything after it back by one place. Figure 1 shows both. With period 3, `abcabcabc…` becomes `abcadbcabc…` after an insertion and `abcacabc…` after a deletion. After the glitch the sequence is still locally periodic, but its phase has moved. A single glitch therefore changes f on about half of the domain, so perturbation bounds that count changed entries are useless (research prompt, Approach 2).

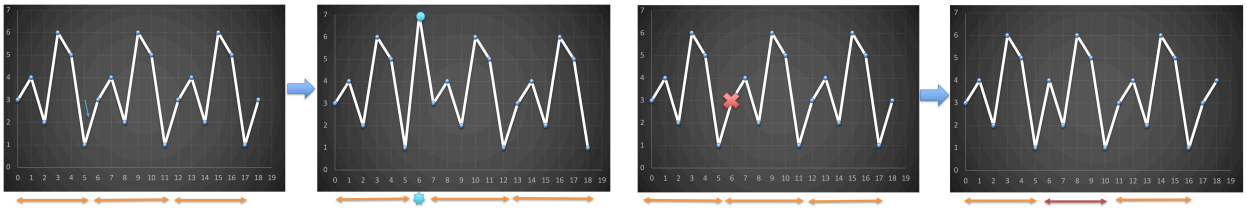


Figure 1: Poster figures 1–2. A periodic function before and after one insertion (left pair) and one deletion (right pair).

3.2 The block-shift model used in 2019

Definition 1 (Block-shift model, "Model I"). from 2019 work Assume Eq. (1). A *k-glitch configuration* consists of

- block boundaries $0 = m_0 \le m_1 \le \ldots \le m_k \le m_{k+1} = M$. Glitch j sits between copy $m_j - 1$ and copy m_j of the period;
- signs $\varepsilon_1, \ldots, \varepsilon_k \in \{+1, -1\}$, where $+1$ is an insertion and -1 a deletion.

Put $s_1 = 0$, $s_{j+1} = s_j + \varepsilon_j$ (cumulative shift) and $B_j = \{m_{j-1}, \ldots, m_j - 1\}$ for $j = 1, \ldots, k+1$. Copy $m \in B_j$ of the period starts at $mr + s_j$, so value w_a occupies $X_a = \{a + mr + s_j : 1 \le j \le k+1, m \in B_j\}$, read modulo N. The Model I distribution is

$$P_k(y) = \frac{1}{N^2} \sum_{a=0}^{r-1} \left| \sum_{x \in X_a} e^{2\pi i x y / N} \right|^2 = \frac{r}{N^2} \left| \sum_{j=1}^{k+1} e^{2\pi i y s_j / N} \sum_{m \in B_j} e^{2\pi i m y / M} \right|^2 . \tag{4}$$

The second equality holds because a enters only through the unimodular factor $e^{2\pi i a y / N}$.

Remark. **Correspondence with the code.** Eq. (4) is `distribution` in `glitchy.py`, which is `probability_distr_with_insertion_deletion` from the 2019 scripts. There `ListK` $= (m_0, \ldots, m_{k+1})$, `Shift` $= (s_1, \ldots, s_{k+1})$, and `abs(z)**2/(mb*N)` equals $r|z|^2/N^2$. Every 2019 run used equal signs, so $s_j = \pm(j-1)$: the code picked "all insertions" or "all deletions" at random. For $k < r$ the points of each X_a are distinct modulo N, so $\sum_y P_k(y) = 1$, which `glitchy.py check` confirms.

Remark. **What Model I idealizes.** Take the *literal model*: an infinite r-periodic sequence, k edits at arbitrary positions, arbitrary inserted symbols, and keep the first N symbols. Model I differs from it in three ways. (i) A glitch strictly inside a period, between residues $b-1$ and b, moves only the classes $a \ge b$ of that copy. Model I moves the whole copy. (ii) The inserted symbols are dropped. (iii) Model I keeps exactly M copies of every value and wraps positions outside $[0, N)$ around. The literal model truncates instead. Each effect changes at most $O(k)$ points of each class. Proposition 10 bounds the resulting error. The 2019 materials do not settle several questions raised in the research

prompt. The bounds are meant to hold for every placement: the poster's conjecture is worst-case, and the code enumerated placements exhaustively or at random. Inserted values play no role in Model I. Adjacent glitches are allowed ($m_j = m_{j+1}$ gives an empty block), but two glitches inside one period are not modelled. k counts operations, and the net displacement is $|s_{k+1}| \le k$. Mixed sign patterns are allowed by Eq. (4). The 2019 runs did not simulate them; see Section 7.3.

4 The distribution near good outcomes

Lemma 2 (Exact form near good outcomes). from 2019 work proof below Write $y = cM + h$ with $0 \le c < r$ and $|h| < M/2$, and let $\omega = e^{2\pi i/r}$. Then

$$R_c(h) := \frac{P_k(cM+h)}{P_0(cM)} = \frac{1}{M^2}\left|\sum_{j=1}^{k+1} \omega^{cs_j} e^{2\pi i h s_j/N} D_j(h)\right|^2, \quad D_j(h) := \sum_{m\in B_j} e^{2\pi i m h/M}, \tag{5}$$

and $|D_j(h)| = |\sin(\pi h L_j/M)/\sin(\pi h/M)|$, where $L_j = m_j - m_{j-1}$.

Proof. Substitute $y = cM + h$ into Eq. (4). Then $e^{2\pi i m y/M} = e^{2\pi i m h/M}$ because $mc \in \mathbb{Z}$, and $e^{2\pi i y s_j/N} = \omega^{cs_j} e^{2\pi i h s_j/N}$. Divide by $P_0(cM) = 1/r$ and use $r^2/N^2 = 1/M^2$. □

For a radius $L \ge 0$ define the *neighbourhood ratio* and its worst case:

$$S_c(L) := \sum_{|h|\le L} R_c(h) = \frac{\Pr[Y \text{ within } L \text{ of } cM]}{\Pr_{\text{ideal}}[Y = cM]}, \quad S^*(L) := \min_{0\le c<r} S_c(L). \tag{6}$$

The 2019 choice is $L = k$. The quantity $S_k = \sum_{y\in G_k} P_k(y)$ in the research prompt equals $(1/r)\sum_c S_c(k) \ge S^*(k)$ when $k < M/2$. Except for the explicitly averaged result below, the subsequent neighbourhood bounds are per c and therefore also bound S_k.

4.1 Two literal glitches: a total-mass bound

The total probability near the *set* of ideal outcomes is easier to bound than the probability near *each* ideal outcome. The following theorem answers the former question directly in the literal model; no continuum limit, boundary rounding, or transfer estimate is needed.

Theorem 3 (Two literal glitches, total mass). new in this note proof below Let $r \mid N$, and start from the infinite periodic sequence with distinct values $w_0, \ldots, w_{r-1}$. Perform two arbitrary single-symbol insertions or deletions, with arbitrary inserted values, and restrict the resulting sequence to its first N entries, obtaining f_2. Let $I \le 2$ be the number of surviving inserted entries in this restriction. For the Fourier measurement Y with distribution P_{f_2}, put $G = \{cN/r : 0 \le c < r\}$ and $G_2 = \{y \in \mathbb{Z}_N : \min_{g\in G} d_N(y,g) \le 2\}$, where d_N is circular distance modulo N. Then, uniformly over all edit locations, signs, and inserted values,

$$\Pr[Y \in G_2] \ge \Pr[Y \in G] \ge \frac{r}{3r+I} \ge \frac{r}{3r+2} \ge \frac{1}{5}. \tag{7}$$

A second bound, useful for large N, is

$$\Pr[Y \in G] \ge \frac{(N-I)^2}{3N^2}. \tag{8}$$

In particular, two deletions give at least $1/3$, and for $N \geq 2$ every sign pattern gives at least $(1-2/N)^2/3$. The assumption $r < \sqrt{N}$ is not needed here.

Proof. For any function $f : \mathbb{Z}_N \to W$, count entries by output value and input residue:

$$n_{wb} := |\{x \in \{0, ..., N-1\} : f(x) = w, x \equiv b \pmod r\}|. \tag{9}$$

Orthogonality of the r roots of unity gives the exact identity

$$\begin{aligned} \sum_{c=0}^{r-1} P_f(cN/r) &= \frac{1}{N^2} \sum_{w} \sum_{b=0}^{r-1} \sum_{b'=0}^{r-1} n_{wb} n_{wb'} \sum_{c=0}^{r-1} e^{2\pi i c(b-b')/r} \\ &= \frac{r}{N^2} \sum_{w} \sum_{b=0}^{r-1} n_{wb}^2. \end{aligned} \tag{10}$$

Track the original index t of each retained, noninserted entry at final index x. Between edits the displacement $s = x - t$ is constant. Each insertion or deletion changes it by one, so two edits leave at most three displacement values among the retained entries. This also holds for adjacent edits or an insertion subsequently deleted. If the original entry has value w_a, then $t \equiv a \pmod r$, hence $x \equiv a + s \pmod r$. Thus the noninserted entries occupy at most $3r$ pairs (w, b). Each of the I surviving inserted entries can add at most one further pair, even if its value is entirely new.

The nonnegative counts n_{wb} therefore have support of size at most $3r + I$ and sum to N. Cauchy–Schwarz yields $\sum_{w,b} n_{wb}^2 \geq N^2/(3r + I)$. Substitution in Eq. (10) proves Eq. (7), since $G \subseteq G_2$ and $r \geq 1$.

For Eq. (8), count only noninserted entries, obtaining counts $m_{wb} \leq n_{wb}$. They sum to $N - I$ and occupy at most $3r$ pairs, so $\sum_{w,b} n_{wb}^2 \geq \sum_{w,b} m_{wb}^2 \geq (N-I)^2/(3r)$. □

Remark. **Model I and the distinction from the conjecture.** For two glitches in normalized Model I (in particular, $2 < r$), the same averaging at exact ideal outcomes gives an even simpler bound. Write $\ell_j = (m_j - m_{j-1})/M$. Then Eq. (5) at $h = 0$ implies

$$\Pr_{\text{Model I}}[Y \in G] = \frac{1}{r} \sum_{c=0}^{r-1} S_c(0) = \sum_{b=0}^{r-1} \left(\sum_{j : s_j \equiv b \pmod r} \ell_j \right)^2 \geq \frac{1}{3}. \tag{11}$$

There are at most three nonzero grouped lengths and they sum to 1. More generally, the same argument gives $1/(k+1)$ for k glitches in normalized Model I. In the literal model, replacing $3r$ by $(k+1)r$ in the proof gives $r/((k+1)r + I)$, where $I \leq k$. These bounds concern the sum over *all* c, including uninformative ones such as $c = 0$. They do not imply a lower bound for $S^*(2)$, prove Conjecture 5, or establish period recovery. The distance convention is immaterial to Eq. (7): the exact set G is also contained in the ordinary, non-wrapped distance-2 neighbourhood.

4.2 Continuum limit

Let $u_j = m_j/M \in [0, 1]$, $\ell_j = u_j - u_{j-1}$ and $\alpha = c/r$. Define the *phase function* and its Fourier coefficients

$$\varphi_\alpha(u) = e^{2\pi i \alpha s_j} \quad \text{for } u \in [u_{j-1}, u_j),$$
$$a_h := \int_0^1 \varphi_\alpha(u) e^{2\pi i h u}\,\mathrm{d}u = \sum_{j=1}^{k+1} e^{2\pi i \alpha s_j} e^{i\pi h(u_{j-1}+u_j)} \frac{\sin(\pi h \ell_j)}{\pi h}, \tag{12}$$

where the $h = 0$ term reads ℓ_j. The continuum neighbourhood mass is $S_\alpha^{\mathrm{cont}}(L) = \sum_{|h|\le L} |a_h|^2$.

Lemma 4 (Continuum approximation). new in this note proof below For all c and h, with $\varepsilon_h := \pi|h|(1 + 2k/r)/M$,

$$\left|\sqrt{R_c(h)} - |a_h|\right| \le \varepsilon_h, \quad \left|S_c(L) - S_{c/r}^{\mathrm{cont}}(L)\right| \le \sum_{|h|\le L} (2\varepsilon_h + \varepsilon_h^2) = O(L^2(1 + k/r)/M). \tag{13}$$

Proof. $\sqrt{R_c(h)}$ is the modulus of $A := M^{-1}\sum_j \omega^{cs_j} e^{2\pi i h s_j/N} D_j(h)$. The sum $M^{-1}D_j(h)$ is a left Riemann sum, with step $1/M$, for $\int_{u_{j-1}}^{u_j} e^{2\pi i h u}\,\mathrm{d}u$. The integrand has derivative of size at most $2\pi|h|$, so each cell contributes error at most $\pi|h|/M^2$, and the total over all blocks is at most $\pi|h|/M$. In addition, $\left|e^{2\pi i h s_j/N} - 1\right| \le 2\pi|h|k/N = 2\pi|h|k/(rM)$, and $\sum_j |D_j|/M \le 1$. So $|A - a_h| \le \varepsilon_h$. Finally, $\left||A|^2 - |a_h|^2\right| \le \varepsilon_h(2 + \varepsilon_h)$ because $|a_h| \le 1$. □

4.3 The poster's "gruesome expression"

The poster states the multi-glitch problem with $l = k$ glitches at $0 < x_1 < \ldots < x_l < t$ and $0 < \alpha < 1$. Write $t = 1$, $x_j = u_j$, and take all insertions, so $s_j = j - 1$. Its quantities are the continuum Fourier data of Eq. (12):

$$A_0 = |a_0|^2, \quad A_{hj} = \frac{\sin(\pi h \ell_j)}{\pi h}, \quad S_h = |a_h|^2 + |a_{-h}|^2, \quad S = A_0 + \sum_{h=1}^{l} S_h = S_\alpha^{\mathrm{cont}}(l). \tag{14}$$

Expanding the squares gives the poster's formulas:

$$\begin{aligned} A_0 &= \sum_{j=1}^{l+1} \ell_j^2 + 2 \sum_{1\le j'<j\le l+1} \ell_j \ell_{j'} \cos(2\pi\alpha(j - j')), \\ S_h &= 2\sum_{j=1}^{l+1} A_{hj}^2 + 4 \sum_{1\le j'<j\le l+1} A_{hj} A_{hj'} \cos(2\pi\alpha(j - j')) \cos\big(\pi h(u_{j-1} + u_j - u_{j'-1} - u_{j'})\big). \end{aligned} \tag{15}$$

The second line uses $\cos(\theta + \psi) + \cos(\theta - \psi) = 2\cos\theta\cos\psi$. Three transcription slips on the poster are corrected here. The first sum in A_0 runs over $j = 1, \ldots, l + 1$, not $0, \ldots, l$. The last term in the second cosine is $x_{j'}$, not x_j. The prefactor t in A_{hj} is consistent only with the normalization $t = 1$. For mixed signs, replace $j - j'$ by $s_j - s_{j'}$. All-deletion configurations give the same S, because S is even in α.

Conjecture 5 (Poster, 2019). conjecture from 2019 work For every $k \ge 1$, every $0 < u_1 < \ldots < u_k < 1$, all-insertion or all-deletion signs, and every $\alpha \in [0, 1)$, we have $S_\alpha^{\mathrm{cont}}(k) \ge 1/2$. By Lemma 4 this gives $S^*(k) \ge 1/2 - O(k^2(1 + k/r)/M)$ in Model I.

The poster interprets this as "even in the worst case, the cost to the runtime would scale like the number of glitches". The $2k + 1$ candidates around each outcome must be tried, but only $O(1)$ more samples are needed (Section 9). The target $S_k \ge c/(k + 1)$ in the research prompt is weaker than Conjecture 5 and already follows from the averaged Model I bound in Section 4.1.

5 A Parseval reformulation

Proposition 6 (Energy and jump formula). new in this note proof below Let $\varphi_j := e^{2\pi i\alpha s_j}$ be the value of φ_α on piece j. Set $u_0 := 0$ and define the jumps

$$\Delta_j := \varphi_j - \varphi_{j+1} \quad (1 \le j \le k), \qquad \Delta_0 := \varphi_{k+1} - \varphi_1 \quad (\text{wrap-around at } u = 0 \equiv 1). \tag{16}$$

1. $\sum_{h\in\mathbb{Z}} |a_h|^2 = 1$, so $S_\alpha^{\text{cont}}(L) = 1 - \sum_{|h|>L} |a_h|^2$.
2. For $h \neq 0$, $a_h = (2\pi i h)^{-1} \sum_{j=0}^{k} \Delta_j e^{2\pi i h u_j}$.
3. $|\Delta_j| = 2|\sin \pi\alpha|$ for every glitch $j \ge 1$, and $|\Delta_0| = 2|\sin(\pi\alpha s_{k+1})|$.

Consequently

$$1 - S_\alpha^{\text{cont}}(L) = \frac{1}{4\pi^2} \sum_{|h|>L} \frac{1}{h^2} \left| \sum_{j=0}^{k} \Delta_j e^{2\pi i h u_j} \right|^2. \tag{17}$$

Proof. (1) Parseval's identity for $\varphi_\alpha \in L^2[0,1)$ with $|\varphi_\alpha| = 1$. (2) Integrate each piece, $\int_{u_{j-1}}^{u_j} e^{2\pi i h u}\,\mathrm{d}u = \big(e^{2\pi i h u_j} - e^{2\pi i h u_{j-1}}\big)/(2\pi i h)$, and collect the coefficient of each $e^{2\pi i h u_j}$. Use $e^{2\pi i h\cdot 1} = 1$ to merge the endpoint terms into Δ_0. (3) $\Delta_j = \varphi_j\big(1 - e^{2\pi i\alpha\varepsilon_j}\big)$ with $\varepsilon_j = \pm 1$, and $s_1 = 0$. □

So the multi-glitch problem is purely about step functions. S is the fraction of the energy of a unimodular step function with $k+1$ jumps that lies at frequencies $|h| \le L$. The k glitch jumps have equal size, and there is one wrap-around jump. All destructive interference is in the exponential sum $\sum_j \Delta_j e^{2\pi i h u_j}$ in Eq. (17).

Corollary 7 (Quadratic radius suffices). new in this note proof below $S_\alpha^{\text{cont}}(L) \ge 1 - 2(k+1)^2/(\pi^2 L)$ for every configuration and every α. In particular, $S_\alpha^{\text{cont}}(L) \ge 1/2$ for $L \ge 4(k+1)^2/\pi^2$.

Proof. In Eq. (17) use $\left|\sum_j \Delta_j e^{2\pi i h u_j}\right| \le \sum_j |\Delta_j| \le 2(k+1)$ and $\sum_{|h|>L} h^{-2} \le 2/L$. □

This bound gives nothing at $L = k$, because $2(k+1)^2/(\pi^2 k) > 1$ for every $k \ge 1$. The crude estimate ignores cancellation between jumps. Exploiting that cancellation, for example with large-sieve-type inequalities for well-separated jump positions combined with merging of clustered jumps, is the obvious route to radius $O(k)$.

6 One glitch

Theorem 8 (One glitch). 2019 claim, proof not on file new in this note proof below Let $k = 1$, with the glitch at $u \in (0,1)$, either sign, and any $\alpha \in [0,1)$. In the continuum limit,

$$S_\alpha^{\text{cont}}(1) = 1 - (1 - \cos 2\pi\alpha) D(u), \quad D(u) := 2u(1-u) - \frac{4}{\pi^2}\sin^2(\pi u) \ge 0. \tag{18}$$

Hence

$$S_\alpha^{\mathrm{cont}}(1) \geq g(u) := 1 - 4u(1-u) + \frac{8}{\pi^2}\sin^2(\pi u), \tag{19}$$

with equality when $\alpha = 1/2$, and $g(u) \geq 8/(8+\pi^2) = 0.44768...$ for all u. The sharp constant is $\min_u g(u) = 0.6400$, attained at $u \approx 0.198$ and its mirror $u \approx 0.802$.

Proof. With $\theta = 2\pi\alpha\varepsilon_1$ we have $\Delta_1 = 1 - e^{i\theta}$ and $\Delta_0 = e^{i\theta} - 1$. So $\left|\Delta_1 e^{2\pi i h u} + \Delta_0\right|^2 = \left|1 - e^{i\theta}\right|^2 \left|e^{2\pi i h u} - 1\right|^2 = 8(1-\cos\theta)\sin^2(\pi h u)$. By Eq. (17), $1 - S = 2(1 - \cos\theta)\pi^{-2}\sum_{|h|\geq 2}\sin^2(\pi h u)/h^2$. The classical identity $\sum_{h\geq 1}\sin^2(\pi h u)/h^2 = \pi^2 u(1-u)/2$ for $u \in [0,1]$ gives $\sum_{|h|\geq 2}\sin^2(\pi h u)/h^2 = \pi^2 u(1-u) - 2\sin^2(\pi u)$. Since $\cos\theta = \cos 2\pi\alpha$, Eq. (18) follows.

For $D \geq 0$, use the symmetry $u \leftrightarrow 1-u$ and take $u \leq 1/2$. If $u \leq 1/3$, then $\sin(\pi u) \leq \pi u \leq \pi\sqrt{u(1-u)/2}$. If $1/3 \leq u \leq 1/2$, then $\sin(\pi u) \leq 1 \leq \pi/3 \leq \pi\sqrt{u(1-u)/2}$. Hence the worst α is $1/2$, which gives g.

Jordan's inequality $\sin(\pi u) \geq 2u$ on $[0, 1/2]$ gives $g(u) \geq 1 - 4u + (4 + 32/\pi^2)u^2$. This quadratic is minimized at $u = 2/(4+32/\pi^2)$, where it equals $1 - 4/(4+32/\pi^2) = 8/(8+\pi^2)$. □

Remark. **Relation to the poster.** The poster says the probability is "greater than 0.447" times the no-glitch probability. The constant $8/(8+\pi^2) = 0.44769$ agrees in every printed digit and comes from the natural argument above: reduce to $\alpha = 1/2$, then apply Jordan's inequality. We therefore believe this was the 2019 argument, but the original proof is not in the repository. Jordan's inequality is the only lossy step. The sharp continuum value is 0.6400. An exhaustive search over the glitch block in Model I gives 0.6403 for $(N, r) = (3000, 50)$ and 0.6400 for $(20000, 20)$. The worst case $\alpha = 1/2$ occurs only for even r and $c = r/2$. Odd r does slightly better. The theorem is stated in the continuum limit. In Model I, Lemma 4 gives $S^*(1) \geq 0.4476 - O((1+1/r)/M)$. Since the bound holds for each c separately, the probability of landing within 1 of a good outcome cM with $\gcd(c, r) = 1$ is at least $0.447 \cdot \varphi(r)/r$.

7 Numerical evidence

7.1 The simulator

Command	What it does
`glitchy.py plot`	Plots, for one placement (given or random), the probability near every good outcome relative to ideal: the neighbourhood sum, the exact good outcome, and a heatmap over the offsets $\|h\| \le k$.
`glitchy.py worst`	Searches every placement of k glitches, or a random sample, for any k and sign pattern. Lists the placements with the smallest $\min_c S_c(k)$.
`glitchy.py check`	Reproduces the numerical results and checks the new total-mass bounds in a few seconds.
`Experiment_plots/Log_Plot{2,3,4,5}.txt`	2019 logs: placement (`ListK`), shifts, and the 20 worst placements. The plot images were too large to commit.

Table 1: Code in the repository. `glitchy.py` (2026) replaces the 2019 scripts `Plot.py`, `Find_Worst_Case.py` and `Test.py`, which remain in git history (commit `a9e171b`). It computes the same distribution: on test configurations, including $r \nmid N$, it agrees with the 2019 formula to 10^{-16}. The `Test.py` experiment (adding one extra glitch) had no logged output and was not ported. Good outcomes are the integers nearest cN/r, the same set as the 2019 rule $(yr \bmod N) \le \lfloor r/2 \rfloor$ or $\ge N - \lfloor r/2 \rfloor$. They equal cM when $r \mid N$.

7.2 Worst cases

Log	N	r	M	k	signs	placements	$S^*(k)$	at c/r	continuum
—	3000	50	60	1	ins.	all 59	0.6403	0.50	0.6400
Plot2	3000	50	60	2	ins.	all 1711	0.7723	0.70	0.7720
Plot3	3000	100	30	3	del.	all 3654	0.6136	0.50	0.6119
Plot4	1500	50	30	4	ins.	all 23751[†]	0.6888	0.40	0.6858
Plot5	3000	50	60	5	del.	5063 (grid)	not logged	—	—

Table 2: Worst neighbourhood ratio $S^*(k) = r \cdot \min_c \sum_{|h| \le k} P_k(cM + h)$ in Model I, over all placements with the given signs, and the good outcome c/r where it is attained. The worst placements (block indices $m_1, ..., m_k$) are $(6, 54)$ for Plot2, $(3, 6, 27)$ for Plot3 and $(2, 4, 6, 8)$ for Plot4, whose mirror image $(22, 24, 26, 28)$ ties. The last column evaluates the same configuration in the continuum limit. [†]The 2019 log searched only 1820 grid placements and found 0.7229 at $(3, 24, 25, 28)$. The full search, the $k = 1$ row and the c/r column were computed for this note. Plot3 and Plot4 have $r > \sqrt{N}$, which does not matter when $r \mid N$ but violates the standing assumption. For Plot2 and Plot3 the full search finds the same minima and minimizers as the 2019 logs.

All worst cases stay well above 1/2. The worst good outcome is $c = r/2$ ($\alpha = 1/2$) for $k = 1, 3$, but $\alpha = 0.7$ (equivalently 0.3, since S is even in α) for $k = 2$ and $\alpha = 0.4$ for $k = 4$. So $\alpha = 1/2$ is not always the worst case. The $k = 4$ minimizer $(2, 4, 6, 8)$ packs the glitches at one end, like the family in Section 8.1. The values are not monotone in k: $0.640, 0.772, 0.614, 0.689$ for $k = 1, 2, 3, 4$, though at different N and r. Any induction on k has to account for the radius growing with k.

7.3 Sign patterns

At $\alpha = 1/2$ the phase function is $\varphi_j = e^{i\pi s_j} = (-1)^{j-1}$ whatever the signs, because a shift of ± 1 changes the phase by $e^{\pm i\pi} = -1$. So in the continuum limit $S_{1/2}$ does not depend on the sign pattern; signs matter only for $\alpha \neq 1/2$. Searching all placements with $N = 3000$, $r = 50$ (numerical evidence only) gives:

signs	$++$	$+-$	$+++$	$++-$	$+-+$	$+--$
$S^*(k)$	0.7723	0.7830	0.6083	0.6083	0.6083	0.6083
at c/r	0.70	0.50	0.50	0.50	0.50	0.50

Flipping every sign gives the same values, because S is even in α. In these runs mixed signs are never worse than equal signs. For $k = 3$ the worst case sits at $\alpha = 1/2$, where signs cannot matter.

7.4 Figures from the poster

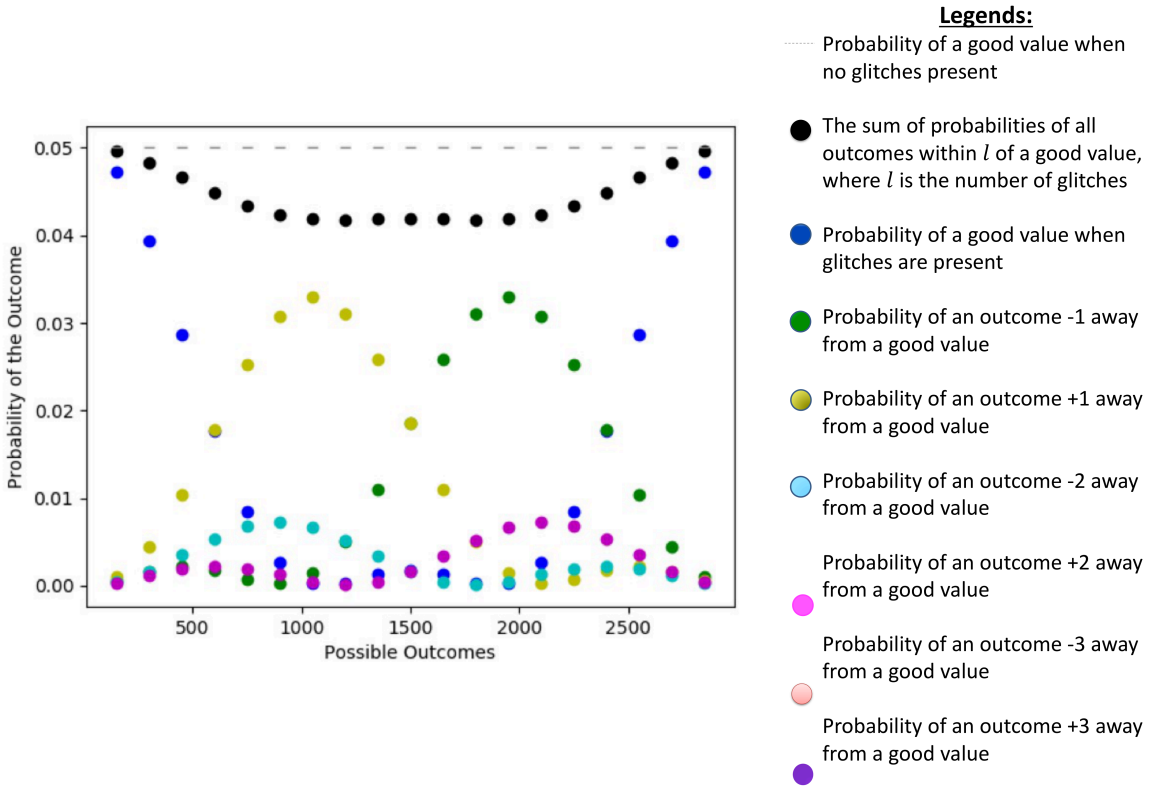


Figure 2: Poster figure 3: $N = 3000$, $r = 20$, $k = 2$. Each column is one good outcome cM. Blue is $P_k(cM)$. Green and yellow are $P_k(cM \mp 1)$, and cyan and magenta are $P_k(cM \mp 2)$. Black is the sum over $|h| \leq 2$. The dotted line is the ideal $1/r = 0.05$. Where the good outcome loses probability, its neighbours gain it.

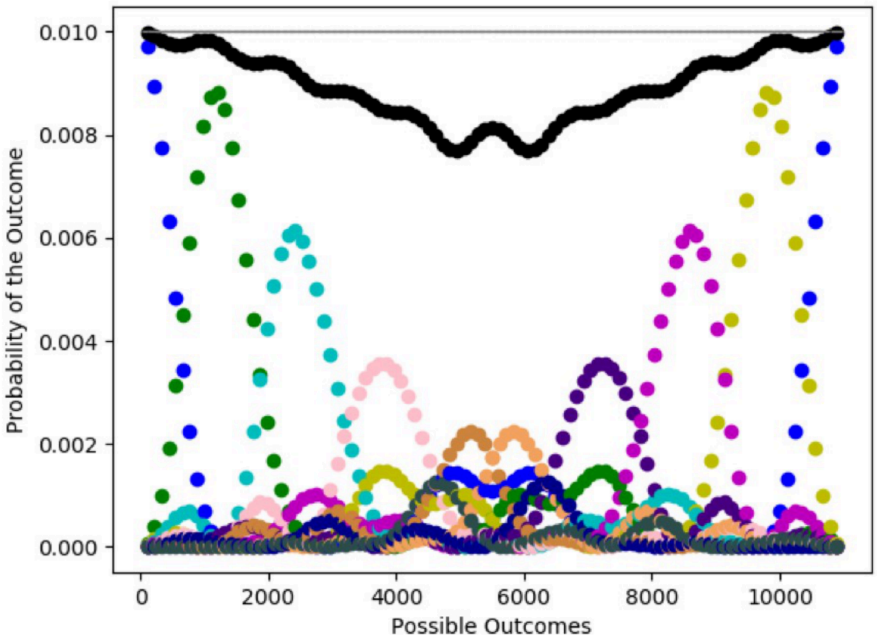


Figure 3: Poster figure 4: $N = 11000$, $r = 100$, $k = 10$, one random placement. The black neighbourhood sum ($|h| \le 10$) stays above about 0.0077, which is 0.77 of the ideal 0.01. The minima are near, but not at, $c = r/2$. The placement was not logged.

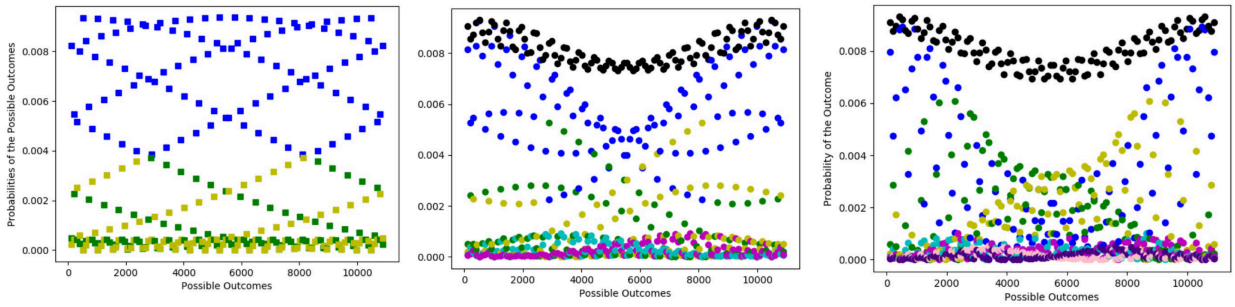


Figure 4: Poster figure 5: $N = 11000$, $r = 107$ ($r \nmid N$) with no glitch (left), 2 glitches (middle) and 3 glitches (right). Without glitches the nearest-integer good outcomes already carry between roughly $4/\pi^2$ and 1 times $1/r$. No analysis exists for this case.

8 Further observations made while compiling this note

8.1 A packed family approaching 1/2

Place all k glitches, with equal signs, equally spaced in $[0, \delta)$, and take $\alpha = 1/2$ (even r, $c = r/2$). Minimizing $S_{1/2}^{\text{cont}}(k)$ over a grid of δ values numerical evidence only gives:

k	2	4	6	10	20	50	100	200
$\min_\delta S$	0.782	0.696	0.653	0.611	0.571	0.542	0.531	0.521
δ	0.245	0.330	0.375	0.415	0.455	0.480	0.490	0.490

Heuristically, on $[0, \delta)$ the phase alternates ± 1 every δ/k. This is a square wave of frequency $k/(2\delta)$, which exceeds k when $\delta < 1/2$. Its energy, about δ, therefore lies outside the window $|h| \le k$, while

the constant part on $[\delta, 1)$ keeps about $1-\delta$. Letting $\delta \uparrow 1/2$ as $k \to \infty$ suggests $\inf S^{\mathrm{cont}}(k) \to 1/2$. **Consequence:** if Conjecture 5 is true, its constant is asymptotically sharp, and a proof must be tight on this family. Neither the limit nor the existence of configurations below $1/2$ has been settled.

8.2 The radius must grow linearly in k

Proposition 9 (Radius lower bound). new in this note proof below Let k be odd, the glitches equally spaced ($u_j = j/(k+1)$), and the signs equal.

1. If $\alpha = 1/2$, then $a_h = 0$ unless h is an odd multiple of $(k+1)/2$. Hence $S^{\mathrm{cont}}_{1/2}(L) = 0$ for $L < (k+1)/2$, and $S^{\mathrm{cont}}_{1/2}(L) = 8/\pi^2$ for $(k+1)/2 \le L < 3(k+1)/2$.
2. If r is odd and $c = (r-1)/2$, which is informative because $\gcd(c, r) = 1$, then $S^{\mathrm{cont}}_{c/r}(L) \le (2L+1)(\pi k/r)^2$ for $L < (k+1)/2$.

Proof. (1) The $k+1$ pieces alternate ± 1, and since $k+1$ is even this holds across the wrap-around too. So $\varphi(u+\tau) = -\varphi(u)$ with $\tau = 1/(k+1)$, and therefore $a_h = -e^{2\pi i h\tau} a_h$. Hence $a_h = 0$ unless $e^{2\pi i h\tau} = -1$. The energy at the fundamental frequency of a ± 1 square wave is $8/\pi^2$. (2) $|a_h(\alpha) - a_h(1/2)| \le \left\|\varphi_\alpha - \varphi_{1/2}\right\|_1 \le \max_j \left|e^{2\pi i(\alpha-1/2)s_j} - 1\right| \le 2\pi|\alpha - 1/2|k = \pi k/r$. Then apply (1). □

So any bound that is uniform in c, or that holds near the informative $c \approx r/2$, needs radius $L \ge (k+1)/2$ (when $r \gg k^{3/2}$). The same configuration also suggests a trade-off. For a general α its energy concentrates near $h \equiv -\alpha(k+1) \pmod{k+1}$, so a radius L captures roughly a fraction $2L/(k+1)$ of the values of c. The total cost, (samples) × (candidates per sample), is then about k for every choice of L. This is heuristic.

8.3 From the literal model to Model I

Proposition 10 (Transfer). new in this note proof sketch only Let f be a literal k-glitch function with $k < r$ (Section 3, remark on idealizations), and round each glitch to the next period boundary to obtain a Model I configuration. Then for all c and L,

$$\left|S^{\mathrm{literal}}_c(L) - S^{\mathrm{Model\ I}}_c(L)\right| = O(Lk(1 + k/M)/M). \tag{20}$$

Together with Lemma 4, a continuum bound $S^{\mathrm{cont}}(k) \ge s_0$ gives $S^{\mathrm{literal}}(k) \ge s_0 - O(k^2(1 + k/r)/M)$. Since $M > \sqrt{N}$, this is meaningful for $k = o(N^{1/4})$.

Sketch. For each class a the two position sets differ in δ_a points (moved, added or removed), with $\delta_a = O(k)$ and $\sum_a \delta_a = O(kr)$. A glitch inside a period moves at most $r-1$ points, one per class. Inserted symbols and truncation contribute $O(k)$ more. New symbols form singleton classes, which add at most k/N^2 to each $P(y)$. Since each class amplitude changes by at most $2\delta_a$ and has size at most $M + \delta_a$, $|P(y) - P'(y)| \le N^{-2} \sum_a 2\delta_a(2M + 2\delta_a) = O(k(M+k)r/N^2)$. Multiplying by $r(2L+1)$ gives the claim. The bookkeeping of δ_a, in particular several glitches in one period and mixed signs, still has to be written out. □

9 From neighbourhood mass to period recovery

The poster's hypothesis is that the period can be learned by examining every point within a certain range of the measured outcome. Made explicit, the procedure is proposal, not proved:

1. Run the period-finding circuit on f_k and measure y.

2. For each y' with $|y' - y| \le L$, expand y'/N in continued fractions and collect the convergent denominators $q \le \sqrt{N}$ as candidates.
3. For each candidate, sample random x and test $f_k(x) = f_k(x+q)$. Output the smallest q whose pass rate exceeds a threshold $1 - \tau$.

Why it should work ($r \mid N$, Model I):

- If y lands within L of cM with $\gcd(c, r) = 1$, then $y' = cM$ is in the window and $y'/N = c/r$ exactly, so $q = r$ is a candidate.
- For $q = r$, the test fails only when a glitch lies in $(x, x + r]$, which happens for an $O(k/M)$ fraction of x.
- For $q \not\equiv 0 \bmod r$, passing requires a glitch in $(x, x + q]$, which happens for at most a $kq/N \le k/\sqrt{N}$ fraction of x. Multiples of r also pass, so the smallest passing candidate is r.
- **Cost.** Each sample needs $2L + 1$ candidates at $\mathrm{poly}(\log N)$ classical cost each. If $S_c(L) \ge s_0$ for every informative c, the expected number of samples is $O(r/(\varphi(r)s_0)) = O(\log\log r/s_0)$. With $L = k$ and Conjecture 5 this is the poster's "cost scales like the number of glitches".

What is missing: a lower bound on the mass near *informative* c, which a bound uniform in c would supply; a careful analysis of the classical test, including inserted symbols; and the $r \nmid N$ case, where candidates must satisfy $|y'/N - c/r| \le 1/(2r^2)$. The averaged guarantee in Theorem 3 does not supply the informative-outcome bound.

10 Open problems

1. **Main conjecture.** Prove Conjecture 5: $S^{\mathrm{cont}}(k) \ge 1/2$. Failing that, prove any constant $c_0 > 0$ at radius $O(k)$, or find a configuration below $1/2$. Known so far: it holds for $k = 1$ (Theorem 8, sharp value 0.640); exhaustive Model I search confirms it for $k = 2, 3, 4$ (Table 2) and for every sign pattern with $k \le 3$ at $N = 3000$, $r = 50$ (Section 7.3); it holds at radius $\Theta(k^2)$ (Corollary 7); and the constant cannot be improved asymptotically (Section 8.1). The step-function form Eq. (17) is the natural setting.
2. **Extremal configurations.** Which α is worst? It is $1/2$ for $k = 1, 3$, but 0.3 for $k = 2$ and 0.4 for $k = 4$ (Table 2). Do worst cases cluster their glitches, as the packed family and the minimizers $(6, 54)$, $(3, 6, 27)$ and $(2, 4, 6, 8)$ suggest? Numerical optimization over $(u_1, ..., u_k)$ in the continuum model for $k \le 10$ is cheap.
3. **Mixed insertions and deletions.** Only the small runs of Section 7.3 exist, and there mixed signs are never worse. Prove that equal signs are always the worst case, or find a counterexample. With mixed signs the phases $\varphi_j = e^{2\pi i \alpha s_j}$ follow a ± 1 walk instead of a geometric progression.
4. **Rigorous transfer.** Complete Proposition 10 with explicit constants, including several glitches in one period, and determine the admissible range of k relative to N and r.
5. **Period recovery.** Turn Section 9 into a theorem. Determine whether averaged bounds such as Theorem 3 can be strengthened to control informative outcomes, and whether the radius–samples trade-off of Proposition 9 is real.
6. $\boldsymbol{r \nmid N}$**.** Extend Lemma 2 to peaks at non-integer positions cN/r, as in Figure 4.
7. **Random glitches.** An average-case model, with uniformly random positions, should be much easier and may give better constants. It is a different theorem from the worst-case one.
8. **Literature.** As of October 2026 the author found no public paper, preprint or manuscript on glitchy period finding beyond the 2019 poster. In particular, nothing is available for the QIP 2026 follow-up mentioned in the research prompt. Do not rely on that follow-up as a source. It remains worth

searching for related work under other names, such as robustness of period finding or of the hidden subgroup problem to oracle errors, and Fourier sampling with corrupted oracles.

9. **Other algorithms.** The poster's future work: robustness of other Fourier-sampling or hidden-subgroup algorithms to glitchy oracles.

A Reproducibility

All code is in one file, `glitchy.py`, at the repository root. It needs `numpy`, and `matplotlib` for plots. With uv the dependencies install automatically:

```
uv run glitchy.py check                                  # every number in this note
uv run glitchy.py plot  --N 11000 --r 100 --k 10         # random placement
uv run glitchy.py plot  --N 3000 --r 50 --blocks 6 54    # given placement
uv run glitchy.py worst --N 1500 --r 50 --k 4            # all placements
uv run glitchy.py worst --N 3000 --r 50 --k 3 --signs 1 -1 1
```

`check` runs in a few seconds. It prints (1) the Model I worst cases of Table 2, recomputed from the logged placements and by full search; (2) the sign-pattern table of Section 7.3; (3) the one-glitch constants of Theorem 8, with the closed form checked against direct summation; (4) the packed-family table of Section 8.1; (5) the radius example of Proposition 9; and (6) checks of the residue-count identity Eq. (10) and the bounds in Theorem 3 against direct Fourier measurement probabilities for literal two-edit sequences. These checks include mixed signs, adjacent edits, repeated or new inserted values, and truncation; the theorem's proof, not the finite checks, supplies the uniform guarantee. The poster figures were cropped from `Summer_Symposium_Poster.pdf`.